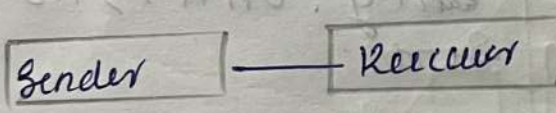
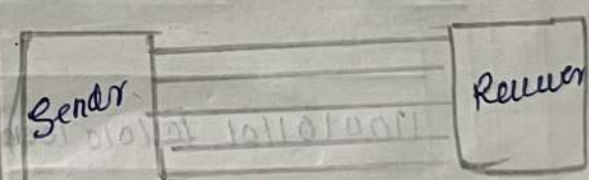


Differentiate b/w Serial and parallel Communication?

Serial Data transfer	parallel Data transfer
Data sent are bit	Data is sent as byte
use a single data time	use multiple wire for transmission.
it more suitable for long distance communication	faster and suitable for short distance communication
only 1 wire occur	multiplexed more occur
eg: USB, RS485	eg: data buses, parallel
 <pre> graph LR S[Sender] --- R[Receiver] </pre>	 <pre> graph LR S[Sender] --- R[Receiver] S --- R S --- R S --- R S --- R </pre>

2 Differentiate b/w Asynchronous Serial Communication and Synchronous Serial Communication?

Synchronous data Communication	Asynchronous data Communication
data is transmitted in block or frame	Data transmitted character by character

Uses a Common Clock / Synchronous
in dr b/w Sender & receiver

Data is transmitted as Continuous Stream

Start & Stop bit or not required each Character

Has higher transmission Speed

eg: SPI, I²C

Sender

Receiver

1100101101 101010 10010

No Common Clock required b/w Sender & receiver.

Data transmitted with gap b/w Character

each Character requires Start & Stop bits

Has comparatively lower transmission Speed

eg: UART, RS232

Sender

Receiver

0100

1010

110

Chars

3 with figure Compare Simplex, half duplex and duplex transmission

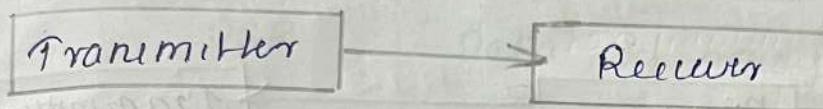
• Simplex transmission:-

* Data flow in one direction only

* one device always the sender and other always receiver.

* No reverse communication is possible
* simple and expensive.

* $A \rightarrow B$



Half - Duplex Transmission: -

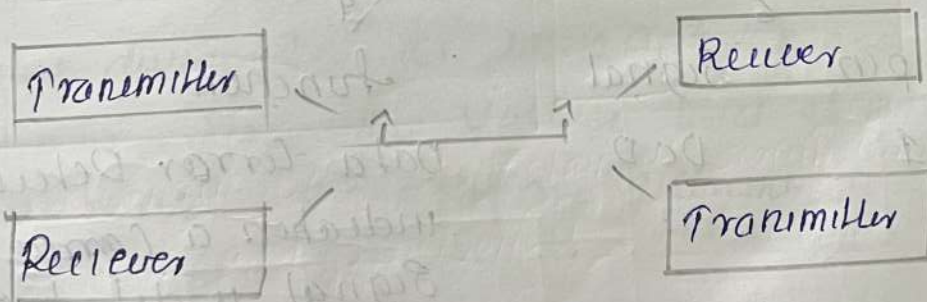
* Data flows in both directions, one at a time

* Both devices can act as sender and receiver

* Communication direction can be changed

* more complex than simplex

* $A \rightarrow B$ or $A \leftarrow B$



Full - Duplex Transmitter: -

* Data flows both directions

* Both devices send and receive

* Communication occurs both d

- same

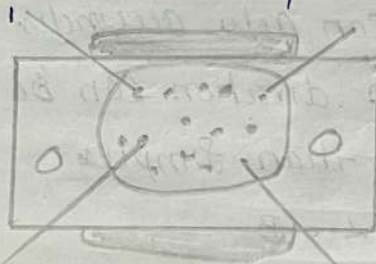
pin package

* more complex and generally require Speechally Communication.

* A → B at same time.



4 Draw DB-9 RS-232 Connector and list the function of the pins.



pin Signal

function

1 DCD

Data-Carrier Detect
Indicates a Carrier
Signal is detected

2 RxD

Receiver Data receive
data

3 TxD

Transmitter Data-Transmit
transmit data

4	DTR	Data Terminal Ready multiplex that the Terminal device ready
5	GND	Signal Ground - reference for signal.
6	DSR	Data Set ready
7	RTS	Data Set Receive
8	CTS	Clear to send transmission
9	RI	Ring indicator

5 Explain ATmega32 Connection to RS232 figure.

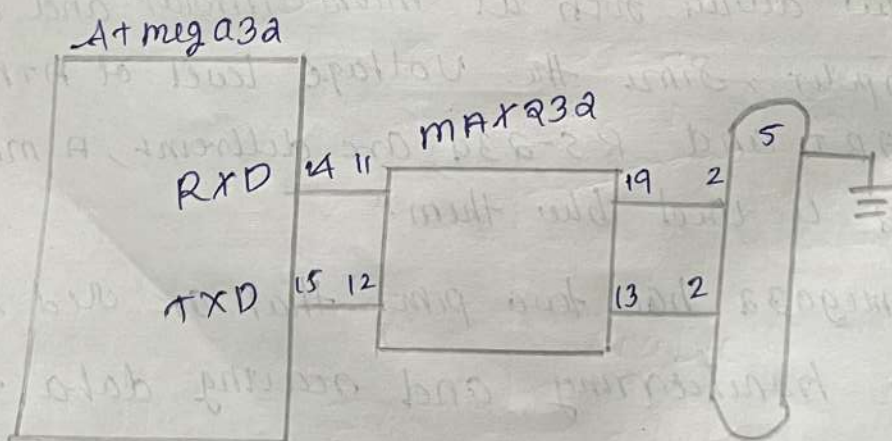
RS-232 is standard serial communication between devices such as microcontroller and a computer. Since the voltage level of ATmega32 UART and RS-232 are different, a MAX 232 is used between them.

ATmega32 has two pins that are used specifically for transferring and receiving data ready. Two pins called TX and RX and are port PORTD Group (PD0 & PD1) for 40-pin package

max232, maxim corp. the max232 convert from RS232 voltage levels to TTL voltage level and vice versa. one advantage of the max232 chip (that uses a 5V power source), which is same at the reverse voltage for the AVR

The line driver used for TX are called T1 and T2 while for RX: R1 & R2. When ATmega32 sends data the signal travel ATmega32 TXD $\rightarrow$ max232 $\rightarrow$ RS-232 RX

Level Converter



40-pin DIP package ATmega32

6

explain LCD interfacing with neat fig?

A LCD (Liquid Crystal Display) is commonly interfacing with the ATmega32 and other microcontroller. The LCD can be operated in 8-bit or 4-bit mode. In 4-bit mode, only four data lines are used.

1. Defining process of LCDs
2. Ability to display number, character & graphs
3. Ease of programming for character & graphs.

LCD pins and their functions:-

VCC, VSS, VEE

While VCC and VSS provide power and ground respectively, VEE is used for controlling LCD contrast.

R_s register select

R_s pin is used for register selection. If $R_s = 0$

instruction command code register is selected

if $R_s = 1$ data register is selected

R/w :-

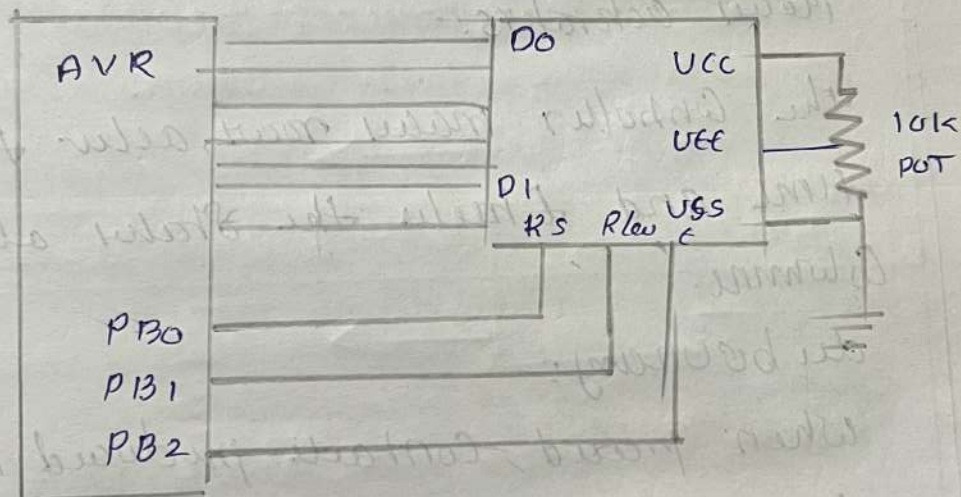
R/w input allow over write information
LCR or read information from R/w 1
reading, R/w=0 when interacting

E - enable :-

latch information provided the data
pins data supported

pin	Symbol	I/O	Description
1	GND	-	Ground
2	VCC		15V power supply
3	VEE		power supply to control circuit
4	RS	I	RS=0 command RS=1 data
5	R/w	I	R/w=0 wait, R/w=1 read
6	E	I/O	enable
7	DB0	I/O	8 bit data bus
8	DB1	I/O	8-bit data bus

9	DB 2	I/O	8-bit databus
10	DB 3	I/O	8-bit databus
11	DB 4	I/O	8-bit databus
12	DB 5	I/O	8-bit databus
13	DB 6	I/O	8-bit databus
14	DB 7	I/O	8-bit databus



7 Explain keyword interfacing with neat figure?

Keyboard is an input device used to enter number, character or command into a microcontroller. A commonly used keyboard with ATmega32 is the 4x4 matrix keypad, which has 16 keys arranged in a 4x4 matrix.

Scanning the micro controller continuously rows and columns to determine which key pressed or not has been pressed regarding its input key pressed. If all rows are generated and key is pressed.

Row drivers:-

The controller makes rows active for a time and checks the status of all columns.

debouncing:-

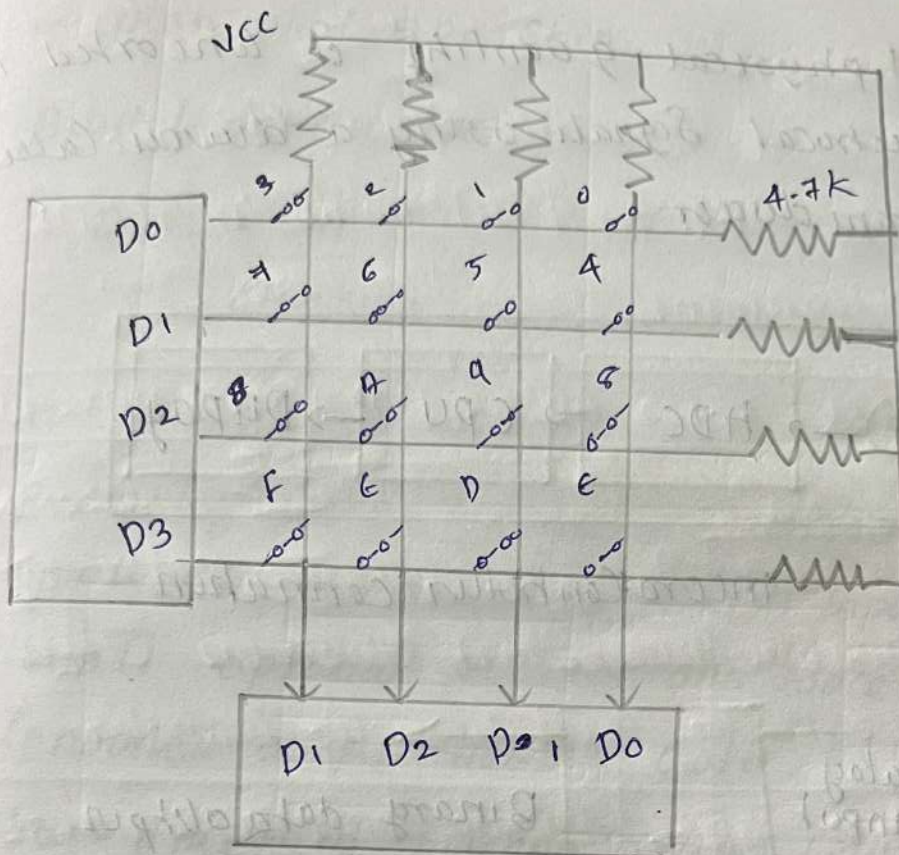
When pressed, contacts produced make transition

Advantage

* Requires fewer I/O ports

* Simple and experienced

* easy to interface with Arduinos

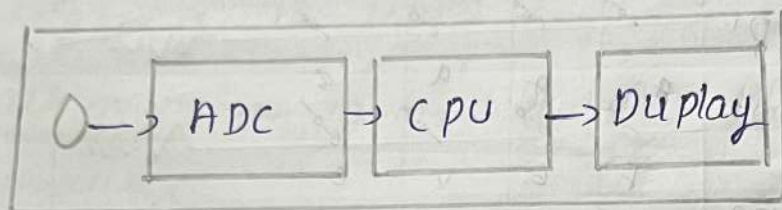


Matrix Keyboard Connection of port B.

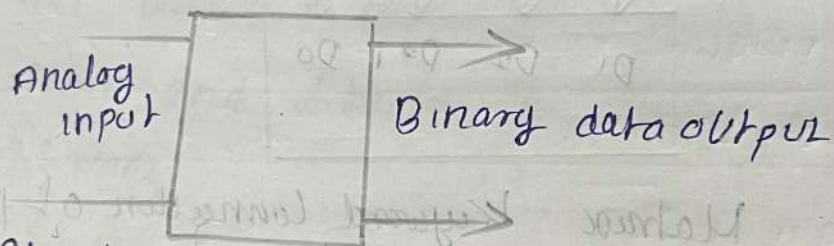
8 Explain ADC interfacing with real figure?

~~2m 31~~ Analog to digital converter is used to convert analog signal such as temperature, light or voltage into a digital interface process by the Atmega32. ADC module widely and data acquisition. Digital computer binary by a physical world

A physical quantity is converted into electrical signals using a device called transducer.



microcontroller connection



Start Conversion

8-bit ADC Block Diagram

Characteristics of ADC :-

Conversion time :-

defined as the time ADC to convert analog input to digital number

V_{ref} :-

Voltage used for reference voltage.
The voltage connected to the pin.

resolution of ADC chip

Digital data output:-

$$D_{out} = \frac{V_m}{\text{Step size}}$$

- (i) 10-bit ADC
- (ii) 8-analog input channel, 7 differential input channel and differential input channel optional of 10x and 200x
- (iii) Converted output having data held by two 32-bit registers called ADC11
- (iv) Because of 16-bit ADC12 register gives 16-bit and ADC ~~12~~ data output only 10-bit inside, 6 bits of 16 are unused.
- (v) internal 2.56V reference or external AREF pin
- (vi) Conversion time is divided by crystal frequency connected to the XTAL pins and ADPSC: 2 bits.

ADMUX :-

Select the ADC input channel Select the Vref and determine its result alignment.

REFS1 REFS0 ADLAR MUX4 MUX3 MUX2 MUX1 MUX0

ADCSRA :-

Control, the ADC operation. It is used to enable ADC Start Conversion, Select prescaler, and multiplier Compilers.

ADEN ADSC ADIF ADIE ADPS2 ADPS1 ADPS0

ADCL = Store 8-bit of 10 bit ADC Conversion

ADCH :- Store upper 2 bit of the 10 bit ADC Conversion

ST=0x12 :- Used to select the ADC - to register value in ATmega32

ADIFSR :- enable ADC interrupt when read

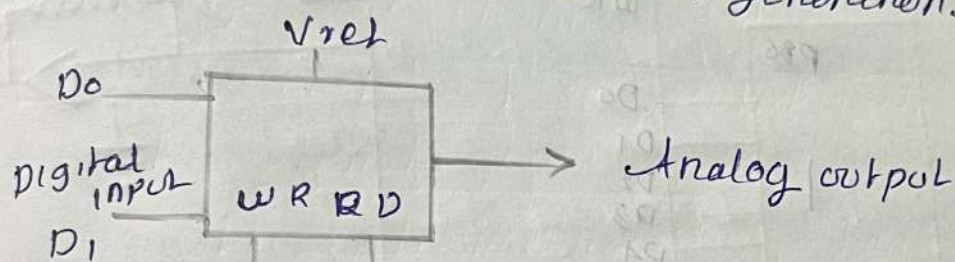
$$ADC \text{ value} = \frac{V_m}{V_{ref}} \times 1023$$

$$V_{ref} = 5V$$

$$0V = 0$$

9 Explain DAC interfacing with neat figure?

MC (Digital to - Analog Converter) is a device that converts digital signal from the microcontroller into an equivalent analog voltage or current. It is useful application such as waveform generation.



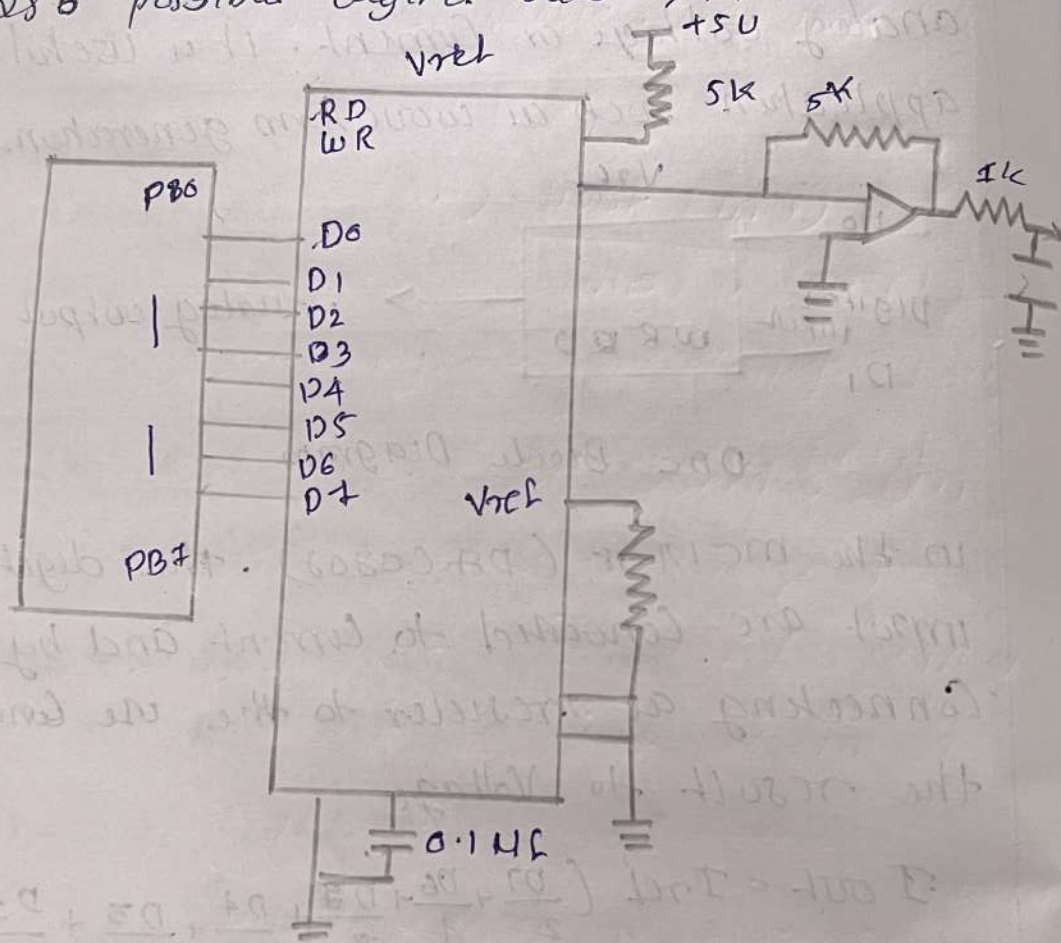
DAC Block Diagram

In the MC1408 (DAC0808). The digital input are converted to current and by connecting a resistor to the I_{out} convert the result to voltage.

$$I_{out} = I_{ref} \left(\frac{D7}{2} + \frac{D6}{4} + \frac{D5}{8} + \frac{D4}{16} + \frac{D3}{32} + \frac{D2}{64} + \frac{D1}{128} + \frac{D0}{256} \right)$$

The Atmega 32 sends a digital value through one of its ports. The DAC converts the digital value into a proportional analog current.

Same DACs also produce a converted output, an op-amp convert to voltage conversion generally to obtain required analog voltage for an 8-bit ADC, then are 256 possible digital value, from 0-255



AVR Connection to DAC0808

10 Explain sensor interfacing with neat figure

A Sensor interfacing:-

2m 34 is a analog temperature sensor. It gives an output voltage directly proportional to temperature in Fahrenheit. It can be connected to the ADC input of an AVR ATmega32 to measure temperature digitally.

Temperature Sensor:

Transducers convert physical such as temperature, light intensity, flow and speed of electrical signal. Temperature, converted to electrical signal, called a thermometer. Resistance temperature change by changing resistance but response not linear.

Temperature

Analog

The output of 2m34 is connected to ADC (P10) of ATmega32.

ARE Conversion:-

ATmega32 is a 10-bit ADC. Converts analog

Voltage LM39 to digital Valve blow
0 and 1023

Voltage measurement:-

The microcontroller calculates to sensor
Voltage from ADC result temperature
-Calculators.

$$\text{Temperature (oh)} = \frac{\text{Vout (mv)}}{10}$$

Signal Conditioning:-

The signal needs to be continuous, we
need to convert signal to voltage, how
over moment send input an concerns
A/D has 10 bit resolution maximum
of 1023 steps and LM39 produce
10mV frequency degree of temperature
change. $2.56 / 1024 = 2.5 \text{mV}$.

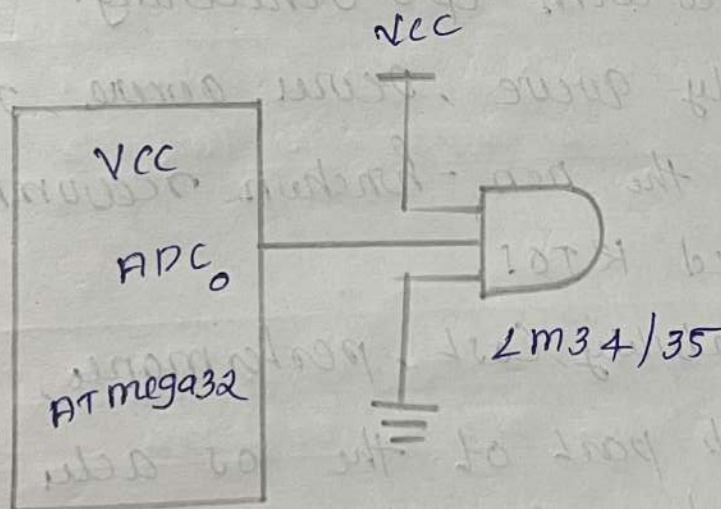
Temp	Vm	# of Step	Bmehtool	Temp. K
0	0	0	00 00000	0000000
1	10	4	00000000	00000001
2	20	8	000010000	00000010

1m34 u Connected to Channel 0

output of A/D converted to get real

temp

To decide the output of AD by
choose left shift option & only read
ADC register



LM34/35 Connection to AVR